



# Chemical Coordination and Integration

You have already learnt that the **neural system provides a point-to-point rapid coordination** among organs. The **neural coordination is fast but short-lived**.

As the **nerve fibres do not innervate all cells of the body** and the **cellular functions need to be continuously regulated**, a special kind of coordination and integration has to be provided. This function is carried out by **hormones**.

The **neural system and the endocrine system jointly coordinate and regulate the physiological functions** in the body.

| Chapter 19 map                                            | What it covers                                                                                                                 |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 19.1 Endocrine Glands and Hormones                        | What a ductless gland and a hormone are                                                                                        |
| 19.2 Human Endocrine System                               | The ten glands in fixed order: hypothalamus, pituitary, pineal, thyroid, parathyroid, thymus, adrenal, pancreas, testis, ovary |
| 19.3 Hormones of Heart, Kidney and Gastrointestinal Tract | Hormone sources that are <b>not</b> endocrine glands                                                                           |
| 19.4 Mechanism of Hormone Action                          | How a hormone acts once it reaches its target                                                                                  |

☆ **[MEMORY AID - not in NCERT]** Two coordinating systems, two speeds: **neural = fast but short-lived and point-to-point**; **endocrine = slower but sustained and broadcast in the blood**. Every fact in this chapter hangs on that contrast.

## 19.1 ENDOCRINE GLANDS AND HORMONES

- **Endocrine glands lack ducts** and are hence called **ductless glands**. Their **secretions are called hormones**.

| Gland type           | Duct    | Where the secretion goes                                 |
|----------------------|---------|----------------------------------------------------------|
| Endocrine (ductless) | Absent  | Released as a <b>hormone</b> , carried in the blood      |
| Exocrine             | Present | Poured through the duct on to a surface or into a cavity |

① **[NOTE]** *Exocrine gland* - the chapter uses the word without defining it (the **pancreas acts as both exocrine and endocrine gland**, and **secretin acts on the exocrine pancreas**), so the contrast is stated here from the chapter's own words only: an **endocrine gland is ductless**, therefore an **exocrine gland is the duct-bearing kind**. Nothing beyond that is claimed.

The **classical definition** of hormone as a chemical produced by endocrine glands and **released into the blood** and **transported to a distantly located target organ** has a **current scientific definition** as follows.

- **Hormones are non-nutrient chemicals** which act as **intercellular messengers** and are **produced in trace amounts**.

The **new definition covers a number of new molecules** in addition to the hormones secreted by the **organised endocrine glands**.

- **Invertebrates possess very simple endocrine systems with few hormones**, whereas a **large number of chemicals act as hormones** and provide coordination in the **vertebrates**.
- The **human endocrine system** is described here.
- These hormones **regulate metabolism, growth and development of our organs, the endocrine glands or certain cells**.

## 19.2 HUMAN ENDOCRINE SYSTEM

- The **endocrine glands and hormone producing diffused tissues/cells** located in different parts of our body constitute the **endocrine system**.

**Pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads (testis in males and ovary in females)** are the **organised endocrine bodies** in our body (Figure 19.1). Counting the **hypothalamus**, the endocrine system is composed of **hypothalamus, pituitary and pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads (testis and ovary)**.

In addition to these, some **other organs**, e.g., **gastrointestinal tract, liver, kidney, heart** also produce hormones. A brief account of the **structure and functions of all major endocrine glands and hypothalamus** of the human body is given in the following sections.

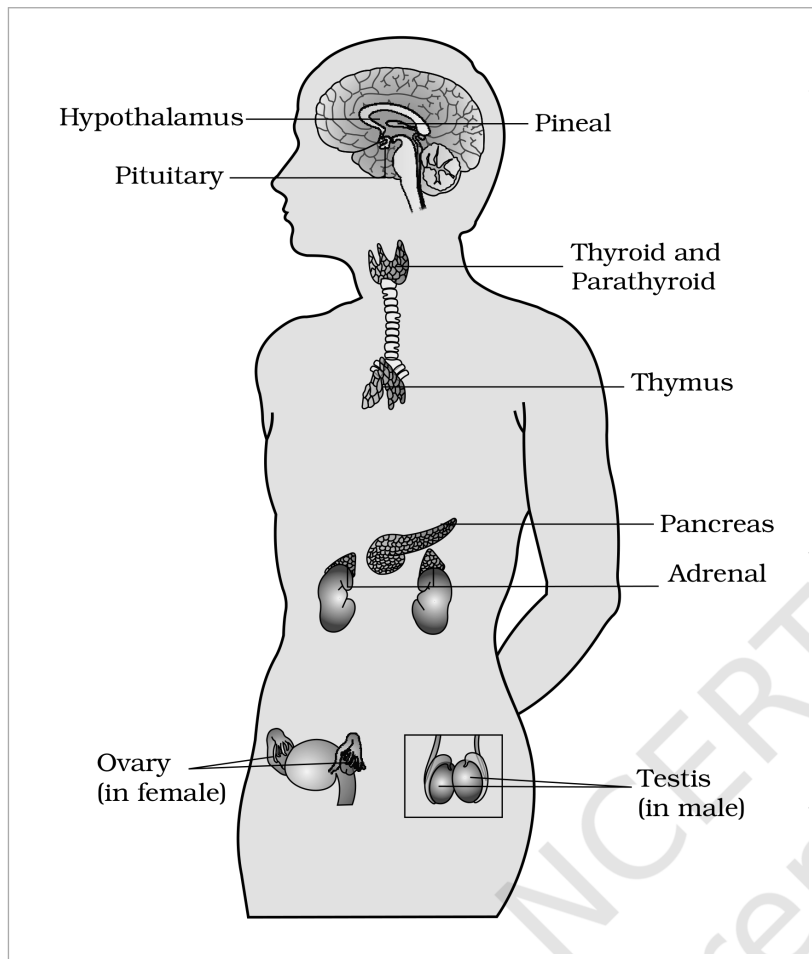


Figure 19.1 Location of endocrine glands

① **[NOTE]** Read the plate (Figure 19.1 labels). Working down the body the plate marks the **Hypothalamus** and, just below it, the **Pituitary**; the **Pineal** on the dorsal side of the forebrain; the **Thyroid and Parathyroid** in the neck; the **Thymus** behind the sternum; the **Pancreas** and, above the kidney, the **Adrenal**; and the gonads - **Ovary (in female)** and **Testis (in male)**.

☆ **[MEMORY AID - not in NCERT]** Ten glands, head to gonad, in the order Section 19.2 uses them: **Hypothalamus, Pituitary, Pineal, Thyroid, Parathyroid, Thymus, Adrenal, Pancreas, Testis, Ovary**. Learn the order once and every subsection number becomes predictable.

### 19.2.1 The Hypothalamus

As you know, the hypothalamus is the **basal part of diencephalon, forebrain** (Figure 19.1) and it **regulates a wide spectrum of body functions**.

- It contains several groups of **neurosecretory cells called nuclei** which produce hormones. These hormones **regulate the synthesis and secretion of pituitary hormones**.

| Hypothalamic hormone type | What it does                              | Example                                                                                                 |
|---------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Releasing hormones        | Stimulate secretion of pituitary hormones | Gonadotrophin releasing hormone (GnRH) stimulates the pituitary synthesis and release of gonadotrophins |
| Inhibiting hormones       | Inhibit secretions of pituitary hormones  | Somatostatin from the hypothalamus inhibits the release of growth hormone from the pituitary            |

- 1 These hormones **originate in the hypothalamic neurons**.
- 2 They **pass through axons** and are **released from their nerve endings**.
- 3 They **reach the pituitary gland through a portal circulatory system**.
- 4 There they **regulate the functions of the anterior pituitary**.

① **[NOTE]** The **posterior pituitary** is under the **direct neural regulation of the hypothalamus** (Figure 19.2) - so the two lobes are reached by two different routes: the **anterior** by **blood** (portal system), the **posterior** by **nerve**.

### 19.2.2 The Pituitary Gland

The pituitary gland is located in a **bony cavity called sella tursica** (the source's own spelling) and is **attached to hypothalamus by a stalk** (Figure 19.2). It is divided anatomically into an **adenohypophysis** and a **neurohypophysis**.

| Anatomical division                        | Part                                                     | Hormones                                                                                                                                                                                                    |
|--------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adenohypophysis - consists of two portions | Pars distalis, commonly called <b>anterior pituitary</b> | <b>Growth hormone (GH), prolactin (PRL), thyroid stimulating hormone (TSH), adrenocorticotrophic hormone (ACTH), luteinizing hormone (LH) and follicle stimulating hormone (FSH) - six trophic hormones</b> |
|                                            | Pars intermedia                                          | Secretes <b>only one hormone</b> called <b>melanocyte stimulating hormone (MSH)</b> ; in humans the pars intermedia is <b>almost merged with pars distalis</b>                                              |
| Neurohypophysis (pars nervosa)             | Also known as <b>posterior pituitary</b>                 | <b>Stores and releases two hormones, oxytocin and vasopressin</b> , which are actually <b>synthesised by the hypothalamus</b> and are <b>transported axonally</b> to neurohypophysis                        |

① **[NOTE]** The pituitary gland is divided into **three major parts**, which are called as **pars distalis, pars intermedia** and **pars nervosa**. Two of the three (distalis, intermedia) make the adenohypophysis; the third is the neurohypophysis.

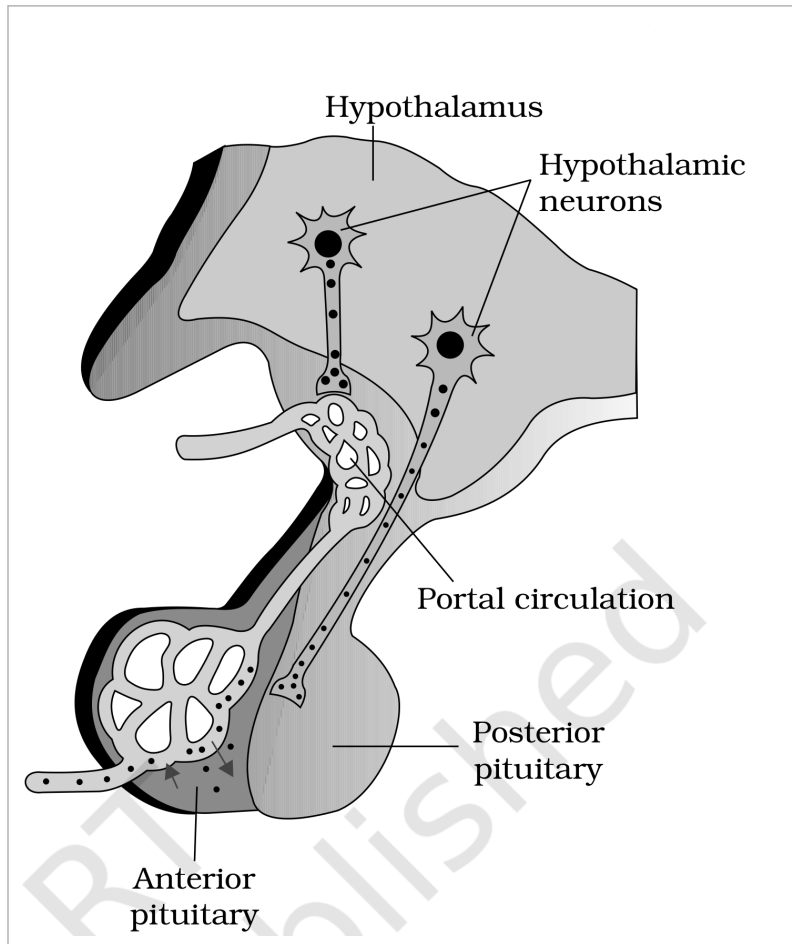


Figure 19.2 Diagrammatic representation of pituitary and its relationship with hypothalamus

① [NOTE] Read the plate (Figure 19.2 labels). The plate marks the **Hypothalamus** at the top with its **Hypothalamic neurons** running down the stalk; the **Portal circulation** carrying hypothalamic hormones into the **Anterior pituitary**; and the **Posterior pituitary**, which the neurons reach directly.

### Growth hormone (GH) - the two directions of error

| Condition          | Secretion                                                              | Effect                                                                                                                                 |
|--------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Gigantism          | Over-secretion of GH                                                   | Stimulates <b>abnormal growth of the body</b>                                                                                          |
| Pituitary dwarfism | Low secretion of GH                                                    | <b>Stunted growth</b>                                                                                                                  |
| Acromegaly         | Excess secretion of growth hormone in adults, especially in middle age | <b>Severe disfigurement (especially of the face)</b> , which may lead to <b>serious complications and premature death if unchecked</b> |

① [NOTE] *Acromegaly is hard to diagnose in the early stages and often goes undetected for many years, until changes in external features become noticeable.*

### What each anterior-pituitary hormone does

| Hormone                                                          | Action                                                                                                                  |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Prolactin (PRL)</b>                                           | Regulates the <b>growth of the mammary glands</b> and <b>formation of milk</b> in them                                  |
| <b>TSH</b> - the exercises call it <b>Thyrotrophin (TSH)</b>     | Stimulates the <b>synthesis and secretion of thyroid hormones</b> from the thyroid gland                                |
| <b>ACTH</b> - the exercises call it <b>Corticotrophin (ACTH)</b> | Stimulates the <b>synthesis and secretion of steroid hormones called glucocorticoids</b> from the <b>adrenal cortex</b> |
| <b>LH and FSH</b>                                                | <b>Stimulate gonadal activity</b> and hence are called <b>gonadotrophins</b>                                            |
| <b>LH in males</b>                                               | Stimulates the <b>synthesis and secretion of hormones called androgens</b> from testis                                  |

| Hormone                                         | Action                                                                                                                                                               |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FSH in males                                    | FSH and androgens regulate spermatogenesis                                                                                                                           |
| LH in females                                   | Induces ovulation of fully mature follicles (graafian follicles) and maintains the corpus luteum, formed from the remnants of the graafian follicles after ovulation |
| FSH in females                                  | Stimulates growth and development of the ovarian follicles                                                                                                           |
| MSH - the exercises call it Melanotrophin (MSH) | Acts on the melanocytes (melanin containing cells) and regulates pigmentation of the skin                                                                            |

### The two posterior-pituitary hormones

| Hormone                                                                                                                                                                                                                                                             | Action                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oxytocin                                                                                                                                                                                                                                                            | Acts on the smooth muscles of our body and stimulates their contraction. In females it stimulates a vigorous contraction of uterus at the time of child birth, and milk ejection from the mammary gland           |
| Vasopressin                                                                                                                                                                                                                                                         | Acts mainly at the kidney and stimulates resorption of water and electrolytes by the distal tubules, thereby reducing loss of water through urine (diuresis); hence it is also called anti-diuretic hormone (ADH) |
| ① [NOTE] An impairment affecting synthesis or release of ADH results in a diminished ability of the kidney to conserve water, leading to water loss and dehydration. This condition is known as <b>Diabetes Insipidus</b> - a water disorder, not a sugar disorder. |                                                                                                                                                                                                                   |

Taken together, the **pituitary hormones regulate the growth and development of somatic tissues** and the **activities of peripheral endocrine glands**.

☆ [MEMORY AID - not in NCERT] Six from pars distalis: GH, PRL, TSH, ACTH, LH, FSH - "Good People Take All Life's Favours". One from pars intermedia: MSH. Two stored in pars nervosa: oxytocin, vasopressin. 6 + 1 + 2.

#### 19.2.3 The Pineal Gland

The pineal gland is located on the **dorsal side of forebrain**. Pineal secretes a hormone called **melatonin**.

- Melatonin plays a **very important role in the regulation of a 24-hour (diurnal) rhythm** of our body.
  - For example, it helps in **maintaining the normal rhythms of sleep-wake cycle, body temperature**.
- In addition, melatonin also influences **metabolism, pigmentation, the menstrual cycle** as well as our **defense capability**.

#### 19.2.4 Thyroid Gland

The thyroid gland is composed of **two lobes** which are located on **either side of the trachea** (Figure 19.3 a). Both the lobes are **interconnected with a thin flap of connective tissue called isthmus**.

- The thyroid gland is composed of **follicles and stromal tissues**. Each **thyroid follicle** is composed of **follicular cells, enclosing a cavity**.

These **follicular cells synthesise two hormones, tetraiodothyronine or thyroxine ( $T_4$ ) and triiodothyronine ( $T_3$ )**. **Iodine is essential for the normal rate of hormone synthesis** in the thyroid.

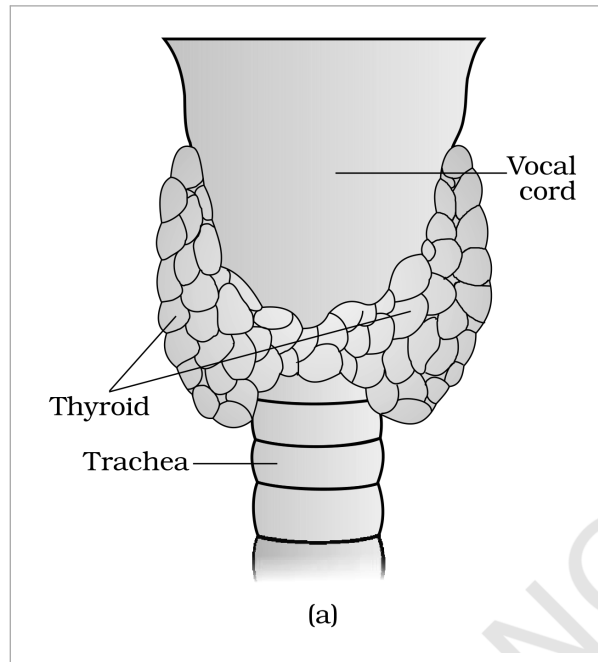


Figure 19.3 Diagrammatic view of the position of Thyroid and Parathyroid (a) Ventral side

① [NOTE] Read the plate (Figure 19.3 (a) labels). On the ventral side the plate marks the **Vocal cord** above, the two lobes of the **Thyroid** astride the **Trachea**.

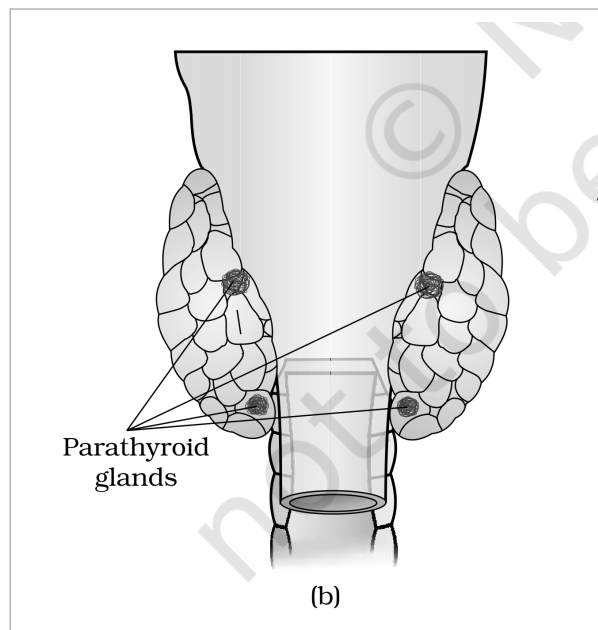


Figure 19.3 Diagrammatic view of the position of Thyroid and Parathyroid (b) Dorsal side

① [NOTE] Read the plate (Figure 19.3 (b) labels). The dorsal view marks the four **Parathyroid glands** on the back of the thyroid lobes - the same organ seen from behind, which is why Section 19.2.5 refers to this panel.

### Too little iodine, too much hormone

| Disorder                          | Cause                                   | Consequences                                                                                                                                                                                                                   |
|-----------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goitre with hypothyroidism</b> | <b>Deficiency of iodine in our diet</b> | Hypothyroidism and <b>enlargement of the thyroid gland</b> , commonly called <b>goitre</b>                                                                                                                                     |
| <b>Cretinism</b>                  | <b>Hypothyroidism during pregnancy</b>  | <b>Defective development and maturation of the growing baby</b> leading to <b>stunted growth (cretinism)</b> , <b>mental retardation</b> , <b>low intelligence quotient</b> , <b>abnormal skin</b> , <b>deaf-mutism</b> , etc. |
| <b>Irregular cycle</b>            | <b>Hypothyroidism in adult women</b>    | The <b>menstrual cycle</b> may become <b>irregular</b>                                                                                                                                                                         |

| Disorder                                                                                   | Cause                                                                              | Consequences                                                                                                                                      |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hyperthyroidism</b>                                                                     | <b>Cancer of the thyroid gland or development of nodules of the thyroid glands</b> | Rate of synthesis and secretion of the thyroid hormones is <b>increased to abnormal high levels</b> , which adversely affects the body physiology |
| <b>Exophthalmic goitre</b> (the source's own spelling), also called <b>Graves' disease</b> | <b>A form of hyperthyroidism</b>                                                   | <b>Enlargement of the thyroid gland, protrusion of the eyeballs, increased basal metabolic rate and weight loss</b>                               |

### What the thyroid hormones do

- Thyroid hormones play an important role in the **regulation of the basal metabolic rate**.
- These hormones also **support the process of red blood cell formation**.
- Thyroid hormones **control the metabolism of carbohydrates, proteins and fats**.
- **Maintenance of water and electrolyte balance** is also influenced by thyroid hormones.
- Collected in one line: the thyroid gland hormones play an important role in the regulation of the **basal metabolic rate, development and maturation of the central neural system, erythropoiesis, metabolism of carbohydrates, proteins and fats, menstrual cycle**.
- Thyroid gland **also secretes a protein hormone called thyrocalcitonin (TCT)** which **regulates the blood calcium levels** - thyrocalcitonin regulates calcium levels in our blood **by decreasing it**.

#### 19.2.5 Parathyroid Gland

In humans, **four parathyroid glands** are present on the **back side of the thyroid gland, one pair each in the two lobes** of the thyroid gland (Figure 19.3 b).

- The parathyroid glands secrete a **peptide hormone called parathyroid hormone (PTH)**. The **secretion of PTH is regulated by the circulating levels of calcium ions**.

- 1 Parathyroid hormone (PTH) **increases the  $\text{Ca}^{2+}$  levels in the blood**.
- 2 PTH **acts on bones** and stimulates the process of **bone resorption (dissolution/demineralisation)**.
- 3 PTH also stimulates **reabsorption of  $\text{Ca}^{2+}$  by the renal tubules**.
- 4 PTH **increases  $\text{Ca}^{2+}$  absorption from the digested food**.

- It is, thus, clear that **PTH is a hypercalcemic hormone**, i.e., it **increases the blood  $\text{Ca}^{2+}$  levels**. Along with TCT, it plays a **significant role in calcium balance** in the body.

☆ **[MEMORY AID - not in NCERT]** Calcium see-saw: **PTH pushes calcium UP** (hypercalcemic: bone, kidney, gut all deliver calcium to the blood); **TCT pulls it DOWN**. Same ion, opposite glands - thyroid down, parathyroid up.

#### 19.2.6 Thymus

The thymus gland is a **lobular structure located between lungs behind sternum on the ventral side of aorta**. The **thymus plays a major role in the development of the immune system**.

- This gland secretes the **peptide hormones called thymosins**.
- Thymosins play a **major role in the differentiation of T-lymphocytes**, which provide **cell-mediated immunity**.
- In addition, thymosins also **promote production of antibodies** to provide **humoral immunity**.
- **Thymus is degenerated in old individuals** resulting in a **decreased production of thymosins**. As a result, the **immune responses of old persons become weak**.

#### 19.2.7 Adrenal Gland

Our body has **one pair of adrenal glands, one above of each kidney** (Figure 19.4 a). The gland is **composed of two types of tissues**: the **centrally located tissue is called the adrenal medulla**, and **outside this lies the adrenal cortex** (Figure 19.4 b).

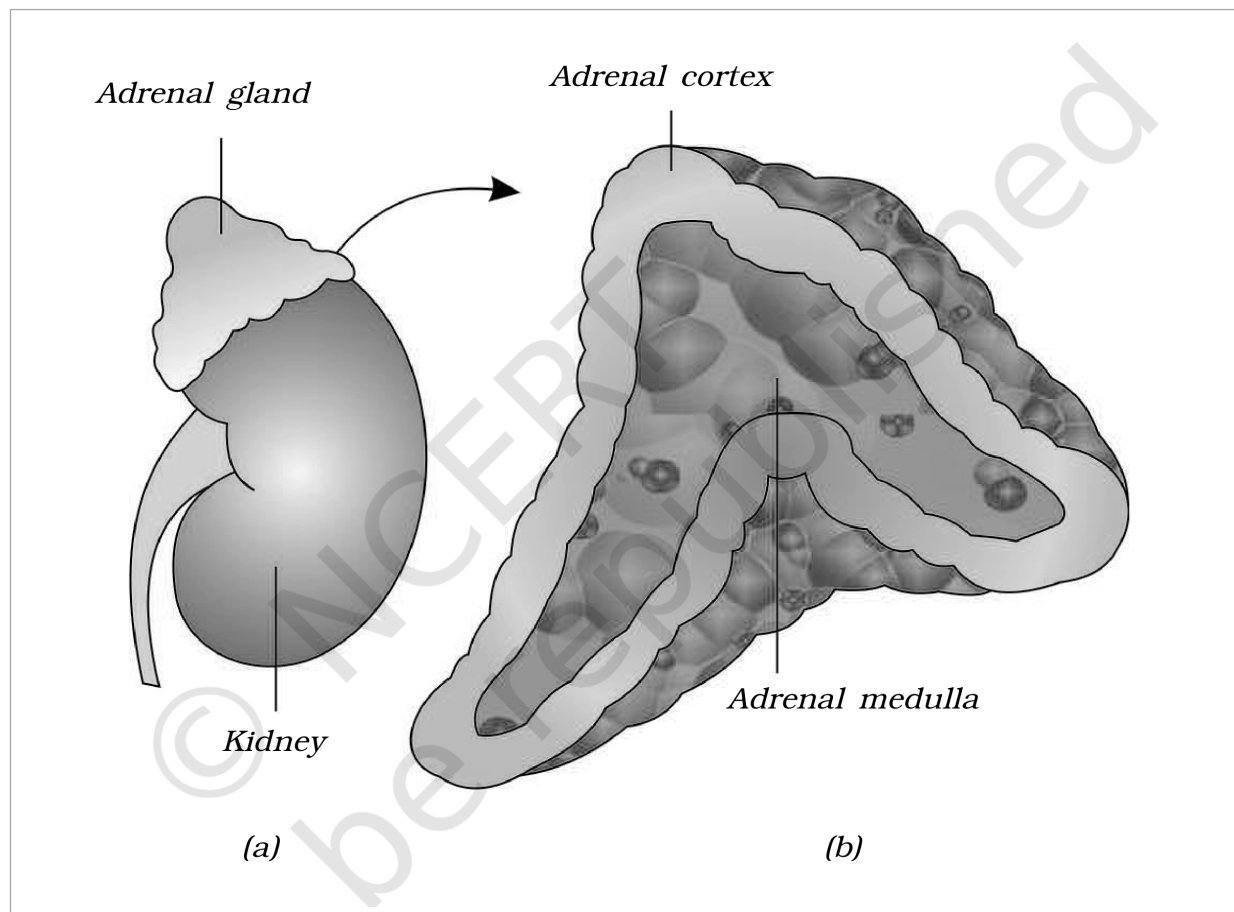


Figure 19.4 Diagrammatic representation of : (a) Adrenal gland above kidney (b) Section showing two parts of adrenal gland

① [NOTE] Read the plate (Figure 19.4 labels). Panel (a) marks the **Adrenal gland** sitting on top of the **Kidney**; panel (b) cuts the same gland open to show the outer **Adrenal cortex** enclosing the central **Adrenal medulla**. Both panels are delivered as one plate because their parts interleave horizontally.

① [NOTE] Underproduction of hormones by the adrenal cortex alters carbohydrate metabolism, causing acute weakness and fatigue, leading to a disease called Addison's disease.

#### Medulla Adrenal medulla - the catecholamines

- The adrenal medulla secretes two hormones called **adrenaline or epinephrine** and **noradrenaline or norepinephrine**. These are commonly called as **catecholamines**.

Adrenaline and noradrenaline are **rapidly secreted in response to stress of any kind and during emergency situations** and are called **emergency hormones** or **hormones of Fight or Flight**.

| Catecholamine effect      | What happens                                                                                                                                    |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Alertness and skin</b> | These hormones increase <b>alertness, pupillary dilation</b> (the source's own spelling), <b>piloerection (raising of hairs), sweating</b> etc. |
| <b>Heart and lungs</b>    | Both the hormones <b>increase the heart beat, the strength of heart contraction</b> and the <b>rate of respiration</b>                          |
| <b>Carbohydrate</b>       | Catecholamines <b>stimulate the breakdown of glycogen</b> resulting in an <b>increased concentration of glucose in blood</b>                    |
| <b>Fat and protein</b>    | In addition, they also <b>stimulate the breakdown of lipids and proteins</b>                                                                    |

① [NOTE] The one-line list to reproduce in an exam: these hormones increase **alertness, pupillary dilation, piloerection, sweating, heart beat, strength of heart contraction, rate of respiration, glycogenolysis, lipolysis, proteolysis**. The last three are the **metabolic names** for the breakdown of glycogen, lipids and proteins above.



The adrenal cortex can be divided into **three layers**, called **zona reticularis (inner layer)**, **zona fasciculata (middle layer)** and **zona glomerulosa (outer layer)**. The adrenal cortex secretes many hormones, commonly called as **corticoids**.

| Corticoid class    | Definition                                                                         | Main one in our body                                         |
|--------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Glucocorticoids    | The corticoids <b>involved in carbohydrate metabolism</b>                          | <b>Cortisol</b> is the main glucocorticoid                   |
| Mineralocorticoids | Corticoids which <b>regulate the balance of water and electrolytes</b> in our body | <b>Aldosterone</b> is the main mineralocorticoid in our body |

- **Glucocorticoids stimulate gluconeogenesis, lipolysis and proteolysis; and inhibit cellular uptake and utilisation of amino acids.**
- **Cortisol** is also involved in **maintaining the cardio-vascular system** as well as the **kidney functions**.
- **Glucocorticoids, particularly cortisol, produce anti-inflammatory reactions and suppress the immune response.**
- **Cortisol stimulates the RBC production.**
- Collected in one line: **glucocorticoids stimulate gluconeogenesis, lipolysis, proteolysis, erythropoiesis, cardio-vascular system, blood pressure, and glomerular filtration rate, and inhibit inflammatory reactions by suppressing the immune response.**

- 1 Aldosterone acts mainly at the renal tubules.
- 2 It stimulates the reabsorption of  $\text{Na}^+$  and water.
- 3 It stimulates the excretion of  $\text{K}^+$  and phosphate ions.
- 4 Thus aldosterone helps in the **maintenance of electrolytes, body fluid volume, osmotic pressure and blood pressure.**

① [NOTE] Small amounts of androgenic steroids are also secreted by the adrenal cortex, which play a role in the growth of axial hair, pubic hair and facial hair during puberty.

☆ [MEMORY AID - not in NCERT] Cortex from outside in - "salt, sugar, sex": **zona glomerulosa** (outer) is the salt layer, **zona fasciculata** (middle) the sugar layer, **zona reticularis** (inner) the sex-steroid layer. NCERT lists them inner-first, so read the list backwards.

Pancreas is a **composite gland** (Figure 19.1) which acts as **both exocrine and endocrine gland**.

- The **endocrine pancreas consists of 'Islets of Langerhans'**. There are about **1 to 2 million Islets of Langerhans** in a normal human pancreas, representing only **1 to 2 per cent of the pancreatic tissue**.

| Islet cell  | Hormone  | Chemical nature                                                                                |
|-------------|----------|------------------------------------------------------------------------------------------------|
| alpha-cells | Glucagon | Peptide hormone; plays an important role in <b>maintaining the normal blood glucose levels</b> |
| beta-cells  | Insulin  | Peptide hormone; plays a major role in the <b>regulation of glucose homeostasis</b>            |

① [NOTE] The source prints these two cell names with the Greek letters **alpha** and **beta** before the word **cells**; they are spelled out here so the page stays a plain-ASCII print file. The **two main types of cells in the Islet of Langerhans** are exactly these two.

### Glucagon - the hyperglycemic hormone

- 1 Glucagon acts **mainly on the liver cells (hepatocytes)**.
- 2 It stimulates **glycogenolysis**, resulting in an **increased blood sugar (hyperglycemia)**.

- 3 In addition, this hormone **stimulates the process of gluconeogenesis**, which also **contributes to hyperglycemia**.
- 4 Glucagon **reduces the cellular glucose uptake and utilisation**.
- 5 Thus, **glucagon is a hyperglycemic hormone**.

#### Insulin - the hypoglycemic hormone

- 1 Insulin acts **mainly on hepatocytes and adipocytes (cells of adipose tissue)**.
  - 2 It **enhances cellular glucose uptake and utilisation**.
  - 3 As a result, there is a **rapid movement of glucose from blood to hepatocytes and adipocytes**, resulting in **decreased blood glucose levels (hypoglycemia)**.
  - 4 Insulin also **stimulates conversion of glucose to glycogen (glycogenesis)** in the target cells.
- The **glucose homeostasis in blood** is thus **maintained jointly by the two - insulin and glucagons** (the source's own plural spelling of glucagon).

① *[NOTE] Prolonged hyperglycemia leads to a complex disorder called **diabetes mellitus**, which is associated with loss of glucose through urine and formation of harmful compounds known as **ketone bodies**. Insulin deficiency and/or insulin resistance result in this disease, and **diabetic patients are successfully treated with insulin therapy**.*

☆ *[MEMORY AID - not in NCERT] Alphabet trick: Alpha comes first and gives glucAgon, which raises sugar; Beta gives insulin, which Brings sugar down. Hyper- and hypo-glycemia are therefore alpha and beta respectively.*

19.2.9

#### Testis

A **pair of testis** is present in the **scrotal sac (outside abdomen)** of male individuals (Figure 19.1). **Testis performs dual functions as a primary sex organ as well as an endocrine gland**.

- Testis is composed of **seminiferous tubules and stromal or interstitial tissue**. The **Leydig cells or interstitial cells**, which are present in the **intertubular spaces**, produce a group of hormones called **androgens**, mainly **testosterone**.
- Androgens **regulate the development, maturation and functions of the male accessory sex organs** like **epididymis, vas deferens, seminal vesicles, prostate gland, urethra** etc.
- These hormones **stimulate muscular growth, growth of facial and axillary hair, aggressiveness, low pitch of voice** etc.
- Androgens play a **major stimulatory role in the process of spermatogenesis (formation of spermatozoa)**.
- Androgens **act on the central neural system and influence the male sexual behaviour (libido)**.
- These hormones produce **anabolic (synthetic) effects on protein and carbohydrate metabolism**.
- Collected in one line: the testis secretes androgens, which stimulate the **development, maturation and functions of the male accessory sex organs, appearance of the male secondary sex characters, spermatogenesis, male sexual behaviour, anabolic pathways and erythropoiesis**.

19.2.10

#### Ovary

Females have a **pair of ovaries located in the abdomen** (Figure 19.1). **Ovary is the primary female sex organ which produces one ovum during each menstrual cycle**. In addition, ovary also produces **two groups of steroid hormones called estrogen and progesterone**.

- Ovary is composed of **ovarian follicles and stromal tissues**. The **estrogen is synthesised and secreted mainly by the growing ovarian follicles**.



(cycle - last step feeds back to step 1)



1 A **follicle grows** and secretes **estrogen**.



2 **Ovulation** occurs.

3

After ovulation, the **ruptured follicle** is converted to a structure called **corpus luteum**.

4

The corpus luteum **secretes mainly progesterone**.

| Hormone                                                                       | Actions                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Estrogens</b>                                                              | Produce <b>wide ranging actions</b> such as <b>stimulation of growth and activities of female secondary sex organs</b> , <b>development of growing ovarian follicles</b> , <b>appearance of female secondary sex characters</b> (e.g., <b>high pitch of voice</b> , etc.) and <b>mammary gland development</b> . Estrogens also <b>regulate female sexual behaviour</b> |
| <b>Progesterone</b> - the exercises call it the <b>Progestational hormone</b> | <b>Progesterone supports pregnancy</b> . It also <b>acts on the mammary glands</b> and <b>stimulates the formation of alveoli</b> (sac-like structures which store milk) and <b>milk secretion</b> . Progesterone thus plays a major role in the <b>maintenance of pregnancy</b> as well as in <b>mammary gland development and lactation</b>                           |

19.3

### HORMONES OF HEART, KIDNEY AND GASTROINTESTINAL TRACT

Now you know about the **endocrine glands and their hormones**. However, as mentioned earlier, **hormones are also secreted by some tissues which are not endocrine glands**.

| Non-gland source                                 | Hormone                                                                          | Action                                              |
|--------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Heart</b> - the <b>atrial wall</b>            | <b>Atrial natriuretic factor (ANF)</b> , a very important <b>peptide hormone</b> | <b>Decreases blood pressure</b>                     |
| <b>Kidney</b> - the <b>juxtaglomerular cells</b> | <b>Erythropoietin</b> , a <b>peptide hormone</b>                                 | <b>Stimulates erythropoiesis</b> (formation of RBC) |

1

**Blood pressure is increased.**

2

**ANF is secreted.**

3

This causes **dilation of the blood vessels**.

4

**This reduces the blood pressure.**

- **Endocrine cells present in different parts of the gastro-intestinal tract secrete four major peptide hormones, namely gastrin, secretin, cholecystokinin (CCK) and gastric inhibitory peptide (GIP).**

| GI hormone                                                                                                                                                                                        | Acts on                               | Action                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------|
| <b>Gastrin</b>                                                                                                                                                                                    | The <b>gastric glands</b>             | <b>Stimulates the secretion of hydrochloric acid and pepsinogen</b>                |
| <b>Secretin</b>                                                                                                                                                                                   | The <b>exocrine pancreas</b>          | <b>Stimulates secretion of water and bicarbonate ions</b>                          |
| <b>CCK</b>                                                                                                                                                                                        | <b>Both pancreas and gall bladder</b> | <b>Stimulates the secretion of pancreatic enzymes and bile juice, respectively</b> |
| <b>GIP</b>                                                                                                                                                                                        | The <b>stomach</b>                    | <b>Inhibits gastric secretion and motility</b>                                     |
| ① <b>[NOTE]</b> Taken together, <b>these hormones regulate the secretion of digestive juices and help in digestion</b> - three stimulate a secretion and the fourth, <b>GIP</b> , shuts one down. |                                       |                                                                                    |

**Several other non-endocrine tissues secrete hormones called growth factors.** These factors are **essential for the normal growth of tissues and their repairing/regeneration**.

☆ **[MEMORY AID - not in NCERT]** Four GI hormones, three organs, one inhibitor: Gastrin -> stomach acid, Secretin -> pancreatic bicarbonate, CCK -> pancreatic enzymes + bile, GIP -> the only one that **inhibits**. "GIP is the Grumpy Inhibitory Peptide."

19.4

### MECHANISM OF HORMONE ACTION

Hormones produce their effects on target tissues by **binding to specific proteins called hormone receptors** located in the **target tissues only**.

| Receptor class           | Where it sits                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------|
| Membrane-bound receptors | Hormone receptors present on the <b>cell membrane of the target cells</b>                           |
| Intracellular receptors  | Receptors present <b>inside the target cell</b> , mostly nuclear receptors (present in the nucleus) |

- 1 **Binding of a hormone to its receptor** leads to the **formation of a hormone-receptor complex** (Figure 19.5 a, b).
- 2 Each receptor is **specific to one hormone only** and hence **receptors are specific**.
- 3 **Hormone-Receptor complex formation** leads to certain **biochemical changes** in the target tissue.
- 4 **Target tissue metabolism** and hence **physiological functions are regulated by hormones**.

On the basis of their **chemical nature**, hormones can be divided into groups :

| Group                                      | Examples                                                                         |
|--------------------------------------------|----------------------------------------------------------------------------------|
| (i) peptide, polypeptide, protein hormones | e.g., <b>insulin, glucagon, pituitary hormones, hypothalamic hormones</b> , etc. |
| (ii) steroids                              | e.g., <b>cortisol, testosterone, estradiol and progesterone</b>                  |
| (iii) iodothyronines                       | <b>thyroid hormones</b>                                                          |
| (iv) amino-acid derivatives                | e.g., <b>epinephrine</b>                                                         |

| Receptor used                                                | Does the hormone enter the cell?                                                                                 | What it regulates                                                                                                            |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Membrane-bound                                               | Normally do not enter the target cell, but generate second messengers (e.g., cyclic AMP, $IP_3$ , $Ca^{++}$ etc) | The second messengers in turn <b>regulate cellular metabolism</b> (Figure 19.5a)                                             |
| Intracellular (e.g., steroid hormones, iodothyronines, etc.) | Enter and bind inside the cell                                                                                   | Mostly regulate gene expression or chromosome function by the <b>interaction of hormone-receptor complex with the genome</b> |

① **[NOTE]** Cumulative biochemical actions result in physiological and developmental effects (Figure 19.5b) - which is why a single hormone-receptor binding event can end in tissue growth.

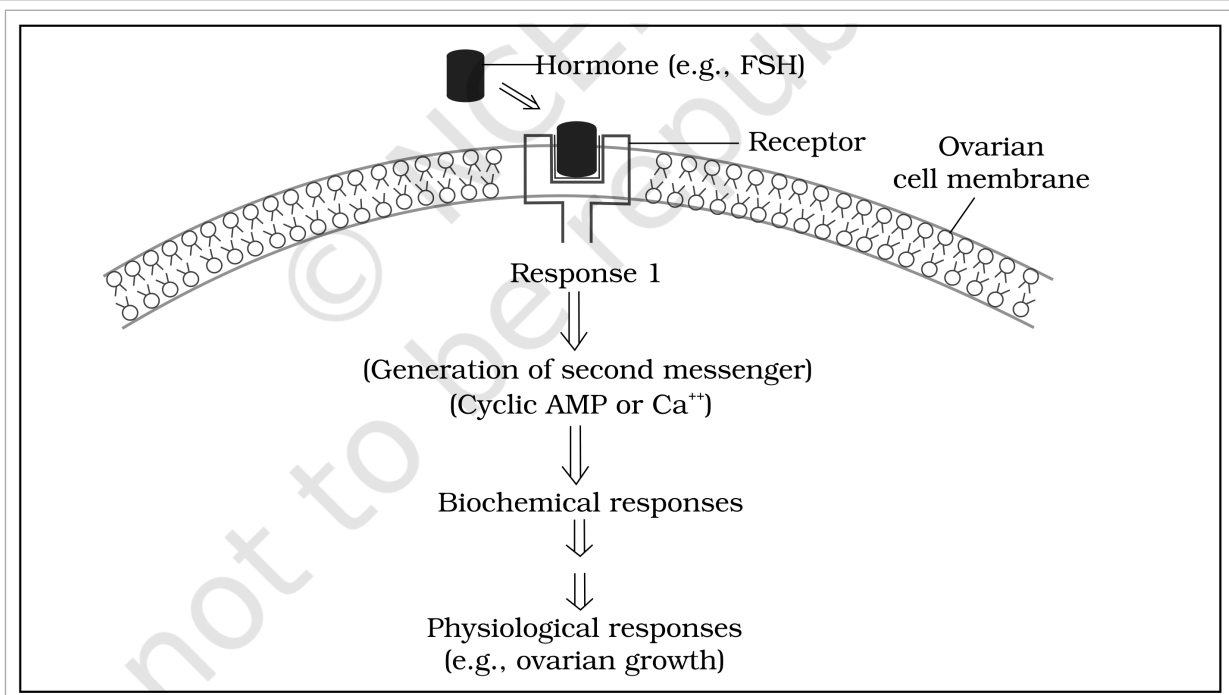


Figure 19.5 Diagrammatic representation of the mechanism of hormone action : (a) Protein hormone

- ① [NOTE] Read the plate (Figure 19.5 (a) labels). A Hormone (e.g., FSH) reaches a Receptor held in the Ovarian cell membrane; that binding is Response 1, the (Generation of second messenger) step (Cyclic AMP or  $\text{Ca}^{++}$ ); the messenger drives Biochemical responses, which end in Physiological responses (e.g., ovarian growth).

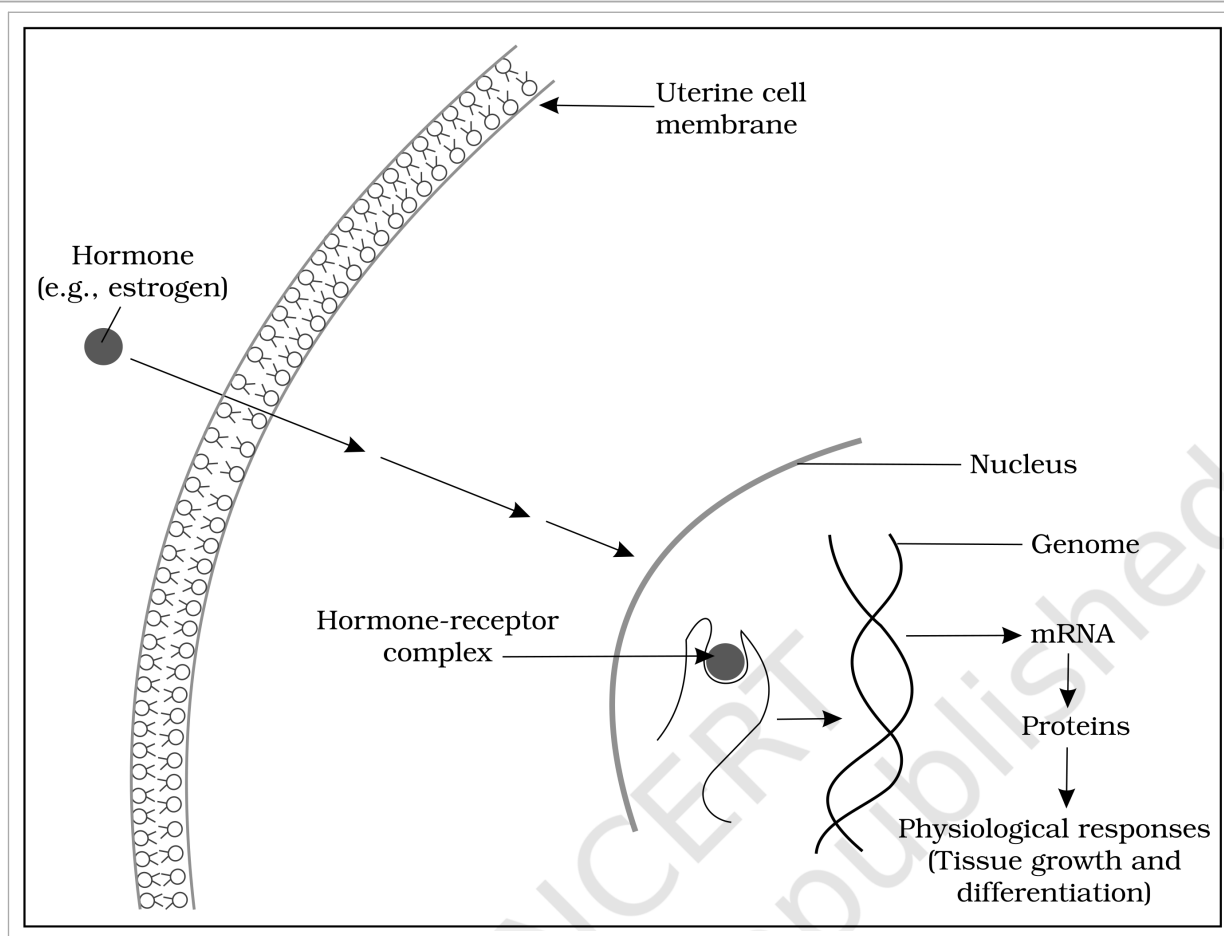


Figure 19.5 Diagrammatic representation of the mechanism of hormone action : (b) Steroid hormone

- ① [NOTE] Read the plate (Figure 19.5 (b) labels). A Hormone (e.g., estrogen) crosses the Uterine cell membrane, and inside the Nucleus the Hormone-receptor complex acts on the Genome; mRNA is transcribed, Proteins are made, and the outcome is Physiological responses (Tissue growth and differentiation).

- ① [NOTE] Both Figure 19.5 captions are printed here with the source's own Diagrammatic, which is how page 11 spells it - Figures 19.2, 19.3 and 19.4 use the correct Diagrammatic in the same book.

- ☆ [MEMORY AID - not in NCERT] Two mechanisms, one question: does the hormone **get in**? Protein/peptide hormones **stay outside** and shout through a **second messenger**; steroids and iodothyronines **walk in** and talk to the **genome**. Water-soluble stays out, lipid-soluble goes in.

#### Recap

#### Quick recap of the whole chapter

- There are **special chemicals which act as hormones** and provide **chemical coordination, integration and regulation** in the human body; they **regulate metabolism, growth and development of our organs**.
- The **endocrine system** is composed of **hypothalamus, pituitary and pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads**; **gastrointestinal tract, kidney and heart** also produce hormones.
- **Pituitary**: pars distalis produces **six trophic hormones**; pars intermedia secretes **only one**; pars nervosa secretes **two**. The pituitary hormones regulate **somatic tissue growth** and **peripheral endocrine glands**.
- **Pineal**: **melatonin** governs the **24-hour (diurnal) rhythm** - sleep and waking, body temperature.
- **Thyroid**: **BMR, central neural system maturation, erythropoiesis, carbohydrate/protein/fat metabolism, menstrual cycle**; **thyrocalcitonin** lowers blood calcium. **Parathyroid**: **PTH** raises it.

- **Thymus:** **thymosins** differentiate **T-lymphocytes** (cell-mediated immunity) and raise **antibody** production (humoral immunity).
- **Adrenal:** **medulla** secretes **epinephrine and norepinephrine**; **cortex** secretes **glucocorticoids** and **mineralocorticoids**.
- **Pancreas:** **glucagon** raises blood glucose (glycogenolysis, gluconeogenesis); **insulin** lowers it (uptake, utilisation, glycogenesis). Failure of insulin gives **diabetes mellitus**.
- **Gonads:** **testis** secretes **androgens**; **ovary** secretes **estrogen and progesterone**.
- **Non-glands:** **ANF** from the heart lowers blood pressure, **erythropoietin** from the kidney drives RBC formation, and the **gastrointestinal tract** secretes **gastrin, secretin, cholecystokinin** and **gastric inhibitory peptide**.

#### Appendix

#### Terms the exercises use, and where they are answered

| Term as the exercises print it                                                                                                                                                             | This chapter's own wording                                                                                                                                                             | Section         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Exocrine gland</b>                                                                                                                                                                      | The chapter defines the <b>endocrine gland as ductless</b> ; the exocrine gland is therefore the <b>duct-bearing</b> kind. <b>Stated as an addition</b> - the chapter never defines it | 19.1            |
| <b>Atrium</b>                                                                                                                                                                              | The body says <b>atrial wall</b> of the heart (ANF)                                                                                                                                    | 19.3            |
| <b>G-I Tract</b>                                                                                                                                                                           | The body says <b>gastro-intestinal tract</b>                                                                                                                                           | 19.3            |
| <b>Hypoglycemic / hyperglycemic hormone</b>                                                                                                                                                | The body uses the nouns <b>hypoglycemia</b> (insulin) and <b>hyperglycemia</b> (glucagon)                                                                                              | 19.2.8          |
| <b>Gonadotrophic hormones</b>                                                                                                                                                              | <b>LH and FSH stimulate gonadal activity</b> and hence are called <b>gonadotrophins</b>                                                                                                | 19.2.2          |
| <b>Blood pressure lowering hormone</b>                                                                                                                                                     | <b>ANF decreases blood pressure</b>                                                                                                                                                    | 19.3            |
| <b>Thyrotrophin, Corticotrophin, Melanotrophin, Progestational hormone</b>                                                                                                                 | <b>TSH, ACTH, MSH and progesterone</b> - the only place the book prints these four names is the exercise list                                                                          | 19.2.2, 19.2.10 |
| <b>Mechanism of action of FSH</b>                                                                                                                                                          | The <b>membrane-bound receptor plus second messenger</b> route, drawn with <b>Hormone (e.g., FSH)</b> in Figure 19.5a                                                                  | 19.4            |
| ① <b>[NOTE]</b> The textbook's <b>final page carries only the word NOTE</b> above a blank panel and the reprint line - it holds <b>no biology</b> , so nothing from it is reproduced here. |                                                                                                                                                                                        |                 |